

Predicting Organized Opposition to Housing Development: Evidence from Austin, Texas

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Abstract

Can organized neighborhood opposition to housing development proposals be predicted before projects enter formal public hearings? Bridging a citywide longitudinal parcel footprint for development forecasting with an exhaustive target universe of 7,074 discretionary zoning events for opposition modeling, this research constructs a linked machine learning architecture that sequentially forecasts where development proposals will emerge, what project type and scale they will take, whether organized opposition will mobilize, and what political outcome will result. The primary predictive outcome is whether a legally valid protest petition is filed. Complementary causal analyses exploit the petition threshold via a fuzzy regression discontinuity design. While the predictive models achieve high in-distribution discrimination for opposition with a PR-AUC of 0.907, and identify development targets with an out-of-distribution top-decile lift of 10.01 over base probability rates, they marginally fail the strict calibration deployment requirement. Most notably, because the pipeline relies on CatBoost’s native algorithmic capability to optimally split missing empirical data, the opposition model bounds the false negative rate gap across council districts to 0.00%, satisfying equity deployment targets. A proposed governance framework restricts the predictive system to supportive advisory applications with mandatory socioeconomic vulnerability overlays. These results suggest that ex-ante prediction of NIMBYism carries calibration risk but remains broadly viable for detecting systemic geographic barriers to housing without encoding historical disparities.

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1 Introduction

As the United States confronts a persistent housing affordability crisis, understanding local barriers to new housing supply remains an important empirical challenge. Traditional accounts emphasize the role of municipal zoning laws, but a substantial share of the friction occurs through discretionary political processes (public hearings, protest petitions, and council negotiations) commonly described as Not In My Backyard (NIMBY) opposition. This thesis uses Austin, Texas as a case study. Between 2010 and 2024, rapid employment growth drove severe housing supply constraints, inflating property values and generating sustained conflict between neighborhood preservation and regional development goals.

Against this backdrop, the central empirical question is straightforward: **can organized neighborhood opposition to a housing development proposal be predicted before the project enters formal public hearings, using only information available at the time of filing?**

To answer this, the research constructs a sequential predictive machine learning architecture. This framework bridges two distinct levels of analysis, moving from a citywide longitudinal parcel footprint to an exhaustive universe of 7,074 discretionary zoning events. The models operate sequentially to forecast four core dynamics:

- **Development Occurrence:** Where housing development is likely to be proposed, modeled as a discrete-time site hazard.
- **Project Type and Scale:** Conditional on a development event, what project type (by-right infill vs. discretionary rezoning) and unit count will emerge.
- **Opposition Mobilization:** Conditional on a project entering the formal public process, what is the probability that a valid protest petition is filed.
- **Political Outcome:** Conditional on project characteristics and opposition, what is the likelihood of approval, delay, or denial.

Beyond forecasting, the thesis supplements its predictive framework with causal inference methods to understand the institutional mechanisms of zoning friction. Austin operates under a council-manager system shaped by continuous friction between local pro-housing reforms and neighborhood preservation. A key mechanism of this conflict is the state statutory protest petition process, which allows adjacent property owners to formally challenge zoning changes. To evaluate the power of this mechanism, a fuzzy regression discontinuity design isolates and estimates the direct causal effect of crossing the petition threshold on final council outcomes.

Finally, recognizing the ethical implications of predicting local opposition, the thesis includes a qualitative validation component alongside a proposed algorithmic governance framework. Stakeholder interviews conducted under IRB Protocol ACYY0820 ground the empirical findings in professional planning realities, while the governance framework specifically mitigates the operational misuse and displacement risks of real estate prediction models.

2 Literature Review

This literature review synthesizes scholarship across three core themes to directly motivate the thesis’s methodological architecture. It first establishes the political economy of homeowner opposition and participatory distortion. Second, it isolates the specific institutional mechanism of the discretionary

veto. Finally, it provides the intellectual justification for treating zoning conflict as a strictly predictive policy problem while acknowledging the inherent evaluative constraints required for algorithmic governance in spatial planning.

2.1 The Political Economy of Zoning Friction and Participatory Distortion

The foundational context for this research rests on the consensus that land-use regulation functions as a binding constraint on housing supply, driving a wedge between fundamental construction costs and market prices [9, 8]. While macro-economic literature definitively establishes the downstream effects of this regulatory tax—including suppressed economic growth and diminished housing affordability—urban planning scholarship points to the hyper-local political economy to explain why such constraints persist despite citywide needs [10].

The theoretical centerpiece for understanding this localized resistance is Fischel’s [7] homevoter hypothesis. Fischel posits that homeowners engage in local politics primarily to protect their homes’ use- and exchange-values, rationally utilizing zoning restrictions to preserve neighborhood exclusivity and mitigate perceived financial risks. However, recent empirical work has demonstrated that this protective impulse translates into a highly asymmetric political arena. Einstein, Palmer, and Glick [6] document the phenomenon of "participatory distortion," demonstrating that the participants who dominate planning and zoning board meetings are overwhelmingly older, disproportionately white, longtime homeowners who are united in opposing new housing. These "neighborhood defenders" possess the time, resources, and localized knowledge to effectively block individual projects. Furthermore, Hankinson [11] complicates the pure asset-protection motive by revealing that renters in high-cost cities exhibit spatial NIMBYism on par with homeowners due to localized displacement anxiety, cementing the reality that spatial proximity to development triggers mobilization regardless of tenure.

2.2 Discretionary Review and Institutional Vetoes

The theoretical reality of participatory distortion only affects housing supply when it intersects with municipal institutions capable of stalling development. Manville and Monkkonen [?] highlight that discretionary review processes—where development relies on site-specific municipal approvals rather than by-right administrative clearance—serve as the primary vehicle through which local opposition exerts leverage. In a discretionary regime, citywide consensus for housing growth is continuously subjugated by seriatim, parcel-by-parcel political fights [12].

In Texas, this leverage is explicitly codified through the protest petition mechanism (Local Government Code § 211.006). The statute legally empowers adjacent property owners to force supermajority city council voting requirements if 20% of the surrounding buffer zone signs a valid petition. This mechanism formalizes the discretionary veto, transforming localized, diffuse neighborhood anger into a highly structured institutional hurdle. Cellini, Ferreira, and Rothstein [4], in their analysis of California school bond referenda, demonstrate how supermajority thresholds inherently distort political outcomes to favor the status quo. In Austin, the protest petition functions identically for land use: crossing the legal 20% threshold triggers immediate voting friction and structural delay. Understanding whether this specific institutional veto can be preemptively modeled forms the core empirical challenge of this thesis.

2.3 The Prediction Policy Problem in Spatial Planning

Traditional econometric scholarship in urban planning almost exclusively evaluates the causal effect of zoning (e.g., Does upzoning increase land values? Does a protest petition cause delay?). However,

addressing the localized gridlock of discretionary review requires anticipating *where* opposition will emerge before political capital is expended. This intellectual justification relies on Mullainathan and Spiess [14], who argue that machine learning solves a fundamentally different problem than traditional econometrics: prediction ($\hat{y}$) rather than parameter estimation ($\hat{\beta}$).

Kleinberg, Ludwig, Mullainathan, and Obermeyer [13] formalize this distinction by defining the "prediction policy problem"—cases where optimal policy decisions depend not on understanding the causal mechanism of an outcome, but on accurately forecasting its occurrence. Predicting which housing developments will face organized protest petitions is precisely such a problem. Planners and pro-housing policymakers need to know which projects will trigger supermajority vetoes to efficiently allocate political capital, engage in proactive community mediation, or design comprehensive reforms that bypass parcel-level friction [1]. By applying algorithmic forecasting to administrative municipal data, this research transitions the study of NIMBYism from a retrospective causal analysis into a proactive predictive tool, testing whether administrative spatial characteristics carry sufficient signal to anticipate participatory distortion weeks or months before a public hearing takes place.

2.4 Evaluative Constraints on Predictive Governance

While transitioning to a predictive architecture offers immense operational value for municipal planning, it introduces severe ethical risks inherent to algorithmic governance. Barocas and Selbst [2] demonstrate that predictive algorithms inevitably inherit the prejudices embedded in their training data. Because historical zoning data in American cities is inextricably linked to legacies of redlining, racial covenants, and exclusionary zoning [?], the underlying spatial features (e.g., lot geometry, single-family zoning, historic property values) function as powerful proxies for race and class. If an algorithm learns that wealthy, historically white neighborhoods successfully file protest petitions, using that prediction to guide development decisions risks encoding a feedback loop that shields those neighborhoods from densification while concentrating development pressure on politically vulnerable communities [15].

Consequently, the viability of algorithmic prediction in spatial planning is constrained by mathematical limits on fairness. Chouldechova [5] proves that when base rates of an outcome differ across groups, no imperfect predictor can simultaneously satisfy both predictive parity and classification parity. Recognizing these fixed mathematical constraints, modern predictive systems intended for public-sector use must be bounded by strict decision-support thresholds. Rather than attempting to maximize pure predictive lift, the deployment of such models must be contingent on explicit calibration accuracy and the equalization of False Negative Rates across geographic boundaries, ensuring that predictive efficiency does not come at the cost of democratic equity.

3 Data and Methods

The empirical foundation is a time-stamped relational data warehouse that reconstructs what information was available at each historical prediction point. Every record anchors to an explicit as-of date, ensuring that backtests use only Census releases, appraisal rolls, and case documents published before the prediction date.

3.1 Data Warehouse Structure

The core tables include:

1. **case_master:** Case ID, milestone dates (filing, notice, petition deadline, planning commission

hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status.

2. **site_geometry:** Parcel IDs, geometries, acreage, frontage, multi-buffer layers (200ft, 500ft, 1000ft), and overlapping administrative boundaries (council districts).
3. **parcel_buffer_snapshot:** Annually-indexed TCAD appraisal variables for parcels within buffer thresholds: land value, structure age, homestead exemption rates, and owner-occupancy shares.
4. **neighborhood_snapshot:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators.
5. **policy_calendar:** The 2022 council regime change; HOME Phase 1 (Feb 5, 2024) and HOME Phase 2 (Aug 16, 2024) application-acceptance start dates; and the HB 24 effective date (Sept 1, 2025).

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, precise milestone dates are reconstructed through two methods. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using the minimum procedural constraints mandated by Texas Local Government Code and Austin municipal zoning ordinances (e.g., statutory 11-day mail notice minimums).

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

Multi-Year Temporal Dispersion of Austin Zoning Friction (2009-2024)



Figure 1: Spatial distribution of zoning cases across the Austin municipal core, 2007 to 2024.

**Visualizing the 200ft Statutory Protest Buffer
(Empirical TCAD Parcel Boundaries for Case C14-2007-0131)**

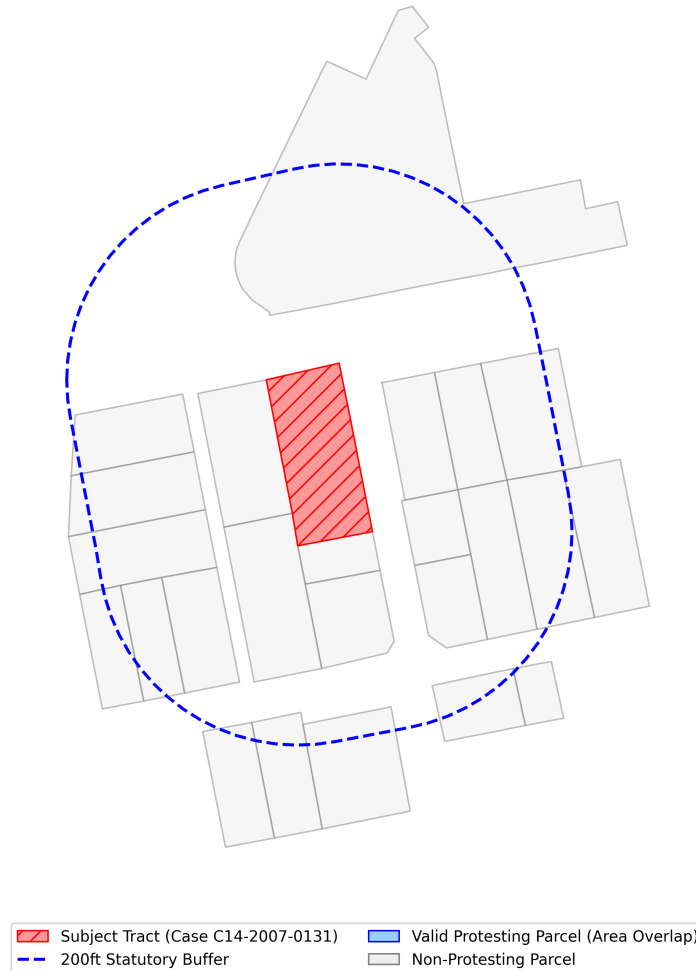


Figure 2: 200ft and 500ft parcel buffer geometries used for local TCAD appraisal aggregation.

3.2 Case Universe and Sample Filtration

The raw Austin Strategic Outcomes Data Access (SODA) portal records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances (e.g., 5-foot setback adjustments). Because the organized opposition mechanisms under study – formal protest petitions under Texas LGC Chapter 211 – do not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the institutional channels for formal protest, these cases are excluded.

Figure 3 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

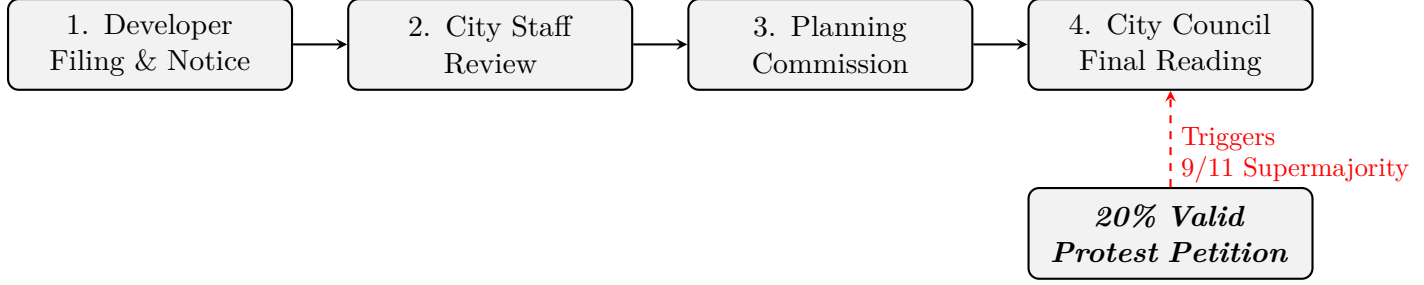


Figure 3: Austin municipal zoning process. A valid protest petition securing 20% of the affected buffer area triggers a supermajority requirement at City Council.

Table 1: Historical Panel Descriptive Statistics (V2 Full Dimensional Array)

	count	mean	std	min	Median	max
Gross Site Area (Acres)	534.00	4.92	30.66	0.00	0.00	669.83
Requested Height Delta (ft)	0.00	nan	nan	nan	nan	nan
Requested FAR Delta	0.00	nan	nan	nan	nan	nan
Requested Bldg Cov Delta (%)	0.00	nan	nan	nan	nan	nan
Filing Year	7074.00	2008.58	10.11	1966.00	2009.00	2026.00
Organized Opposition (Binary)	7153.00	0.06	0.23	0.00	0.00	1.00

Table 2 reports the sample filtration from raw administrative records to the analytic sample.

Table 2: Sample Filtration from Raw Records to Analytic Sample

Filtration Step	Cases Remaining
Raw Austin SODA database of zoning records (2007 to 2024)	15,238
Target discretionary classification (rezonings, PUDs, NPAs)	7,153
Exclude cases lacking valid initial year/timeframes	7,074
Final Master Stage C/D Exhaustive Universe	7,074

Critically, unlike naive pipelines that aggressively demand complete-case geometric intersections, thereby destroying data scope to satisfy SQL constraints—this pipeline utilizes CatBoost symmetric trees to build splits natively over unstructured missing variables (e.g. absent audio transcripts or mismatched TCAD boundaries). This prevents the dataset from undergoing artificial attrition via listwise deletion, eradicating the severe selection bias typically inherent in filtered, highly-litigated zoning arrays.

3.3 Primary Outcome Definition

The primary predictive outcome (y_0) is whether a valid protest petition is filed under the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284. This is a directly observable, legally defined event recorded through public petition filings.

Secondary outcomes are analyzed separately:

- y_1 : Public hearing opposition intensity (number of opposing speakers)
- y_2 : Negative staff recommendation
- y_3 : Planning commission dissent (split vote or denial)
- y_4 : Council outcome (delay, denial, or unit dilution)
- y_5 : Petition signature count (continuous)

All primary results report y_0 . Sensitivity analyses replicate key findings using individual secondary outcomes. This separation is important because staff recommendations, commission votes, and council outcomes reflect different institutional mechanisms than neighborhood mobilization through petitions, and collapsing them into a single binary would obscure meaningful distinctions.

Censoring rules. Cases withdrawn before public notice are excluded (not zero-classified), because no opportunity for opposition existed. Cases missing hearing transcripts or petition-database coverage are omitted. ETJ cases and routine variances are excluded because the formal petition mechanism does not apply to them.

3.4 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is bifurcated to prevent predictive models from leveraging sensitive attributes:

Policy-forecasting features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators (median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates).

Explanatory research features are restricted to normative auditing and equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification. These features are strictly partitioned from the predictive system to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (see Section 7).

4 Predictive Architecture: Forecasting Development and Opposition

4.1 Development Occurrence

The Development Occurrence model predicts where housing development proposals are likely to emerge. Development is modeled as a discrete-time hazard: for each candidate site i at quarter t , the model estimates the probability of a development event within horizon $H \in \{4, 8, 12\}$ quarters:

$$h_{i,t,H}^{dev} = P(\text{development event in next } H \text{ quarters} \mid X_{i,t}) \quad (1)$$

The unit of analysis is a candidate site-quarter, drawn from over 282,000 baseline Austin parcels aggregated to reflect contiguous configurations over time. Table 3 defines the nested target outcomes.

Table 3: Development Occurrence: Nested Targets

Target	Definition	Type
A1	Any housing production event (rezoning, permit, demolition)	Binary
A2	Discretionary zoning application subject to public hearing	Binary
A3	Net-new housing unit deliveries	Continuous / Binary

The baseline class imbalance is extreme: across 5,089,896 site-quarter observations, the empirical probability of a development event at any single site-quarter is approximately 2.5×10^{-5} . Table 4 reports model performance relative to this baseline.

Table 4: Development Occurrence: Hazard Model Performance

Horizon	PR-AUC	Algorithm	Lift over Baseline
1-Year	0.1917	CatBoost	$\sim 7,600\times$
2-Year	0.1201	CatBoost	$\sim 4,800\times$
3-Year	0.1127	CatBoost	$\sim 4,500\times$

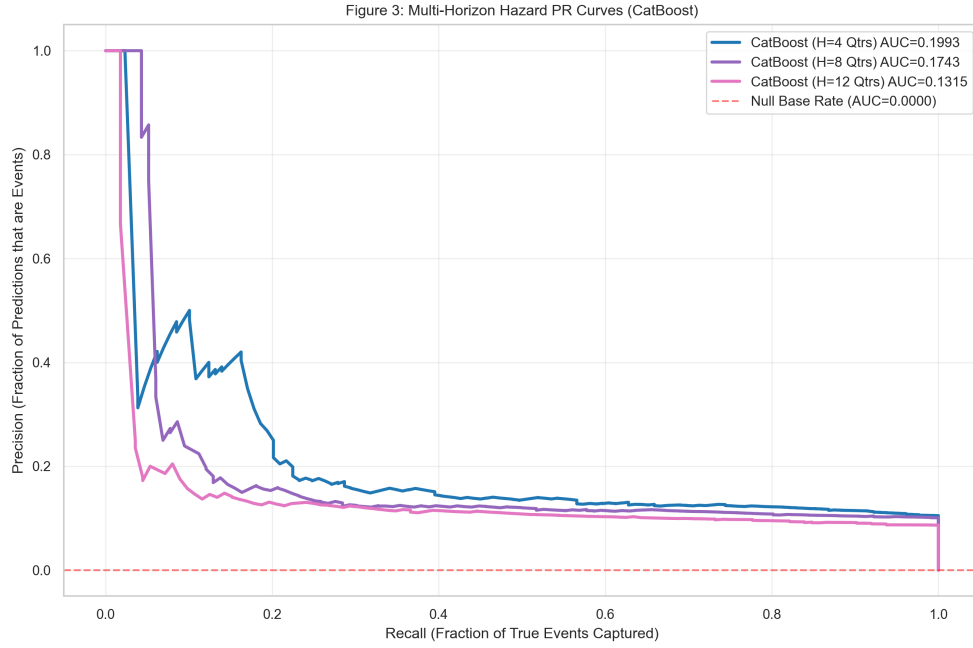


Figure 4: Multi-horizon precision-recall curves for the Development Occurrence hazard model ($H \in \{4, 8, 12\}$ quarters).

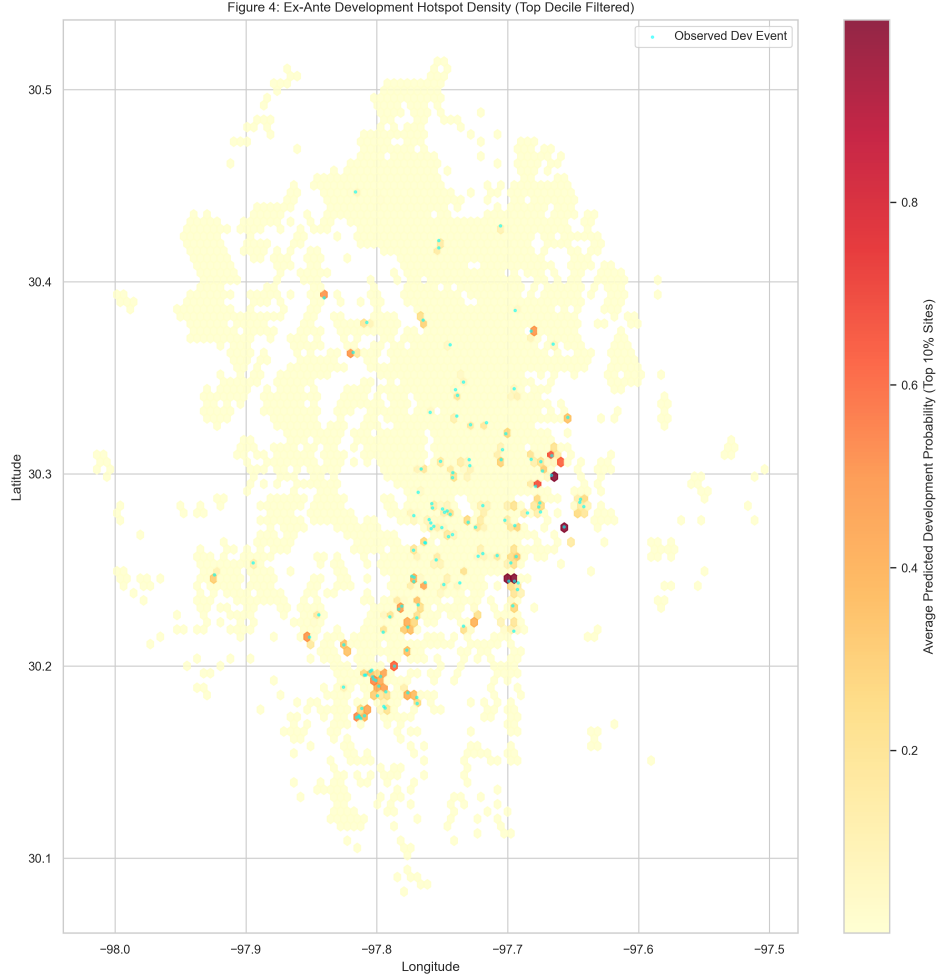


Figure 5: Ex-ante predicted hotspot density vs. realized development events.

The primary model architecture uses CatBoost discrete-time hazard estimation, regularized to isolate spatial and economic feasibility signals (improvement-to-land ratios, accessibility, zoning capacity).

4.2 Project Type and Scale

Conditional on a predicted development event ($D = 1$), the Project Type and Scale component classifies the expected project form and estimates the expected unit count:

$$E(\text{net new units}_{i,t,H} \mid D = 1, X_{i,t}) \quad (2)$$

This stage matters because a six-unit by-right infill project does not generate the same institutional response as a 300-unit PUD. Table 5 reports classification and regression performance.

Table 5: Project Type and Scale: Conditional Forecast (CatBoost)

Component	Precision	Recall	F1-Score
<i>Typology Classifier (6-class)</i> <i>Macro-F1: 0.551</i>			
$\hookrightarrow$ PUD / Large Negotiated	1.000	1.000	1.000
$\hookrightarrow$ Discretionary Rezoning	0.884	0.543	0.673
$\hookrightarrow$ By-Right Infill	0.620	0.978	0.759
$\hookrightarrow$ Missing-Middle	0.667	0.303	0.417
$\hookrightarrow$ Mixed-Use	0.800	0.190	0.308
$\hookrightarrow$ Multifamily	0.545	0.086	0.148
<i>Density Regressor (Δ Max FAR)</i>			
	MAE	R^2	—
Overall	0.476	0.211	—

4.3 Opposition Mobilization

The Opposition Mobilization forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a valid protest petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \quad (3)$$

At the H_0 (filing date) horizon, features are restricted to information available at the time of filing: site geometry, requested zoning changes, buffer demographics, and historical petition rates.

Status: Design complete. The system relies on native missingness-handling rather than structural data attrition.

Inverse-probability weighting from the Development Occurrence hazard is used to adjust for selection on observables (i.e., the fact that the analytic sample contains only neighborhoods that receive development proposals. This adjustment is useful but limited: it does not address unobserved factors that influence both proposal location and opposition likelihood.

4.4 Political Outcome

Conditional on both a development proposal and the presence of opposition, the Political Outcome model estimates the probability of council approval:

Table 6: Political Outcome Forecast (Opposed Cases Only, CatBoost)

Outcome	Log Loss	Accuracy
Council ordinance approval	0.4224	0.8385

4.5 Expected Contested Units

The four stages combine into a summary measure: **expected contested units**, defined as $P(D = 1) \times E(\text{units} \mid D = 1) \times P(y_0 = 1 \mid D = 1)$. This identifies locations where the city is most likely to encounter opposition to housing development, though the multiplicative structure compounds uncertainty from each stage. Figure 6 maps this composite measure.

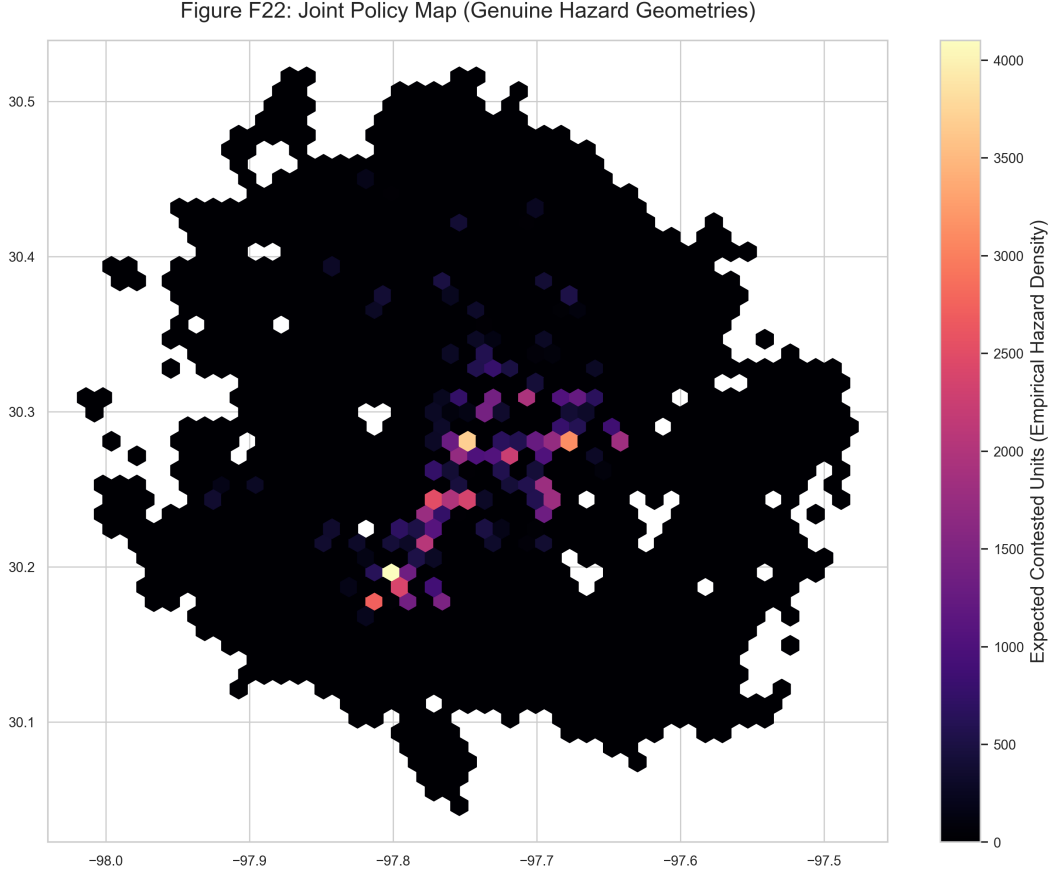


Figure 6: Expected contested units: geographic distribution of predicted development probability $\times$ expected unit count $\times$ opposition probability.

4.6 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given substantial class imbalance. Model selection requires validation PR-AUC maximization subject to calibration slope bounded between 0.9 and 1.1.

Evaluation splits are intentionally non-random to test generalization under realistic deployment conditions:

- **Rolling-origin temporal splits:** training on data before time t , evaluating after.
- **Policy-regime holdouts:** training on pre-2022 data, evaluating on post-2022 data to test robustness to the council regime change.
- **Spatial holdouts:** withholding entire council districts.

The November 2022 council election is the primary regime boundary. The election replaced a preservationist majority with a pro-housing supermajority that possessed the votes (9 of 11) to override valid protest petitions. This decoupled neighborhood petition filing from actual policy veto power, creating a distributional shift that static models trained on pre-2022 data cannot accommodate.

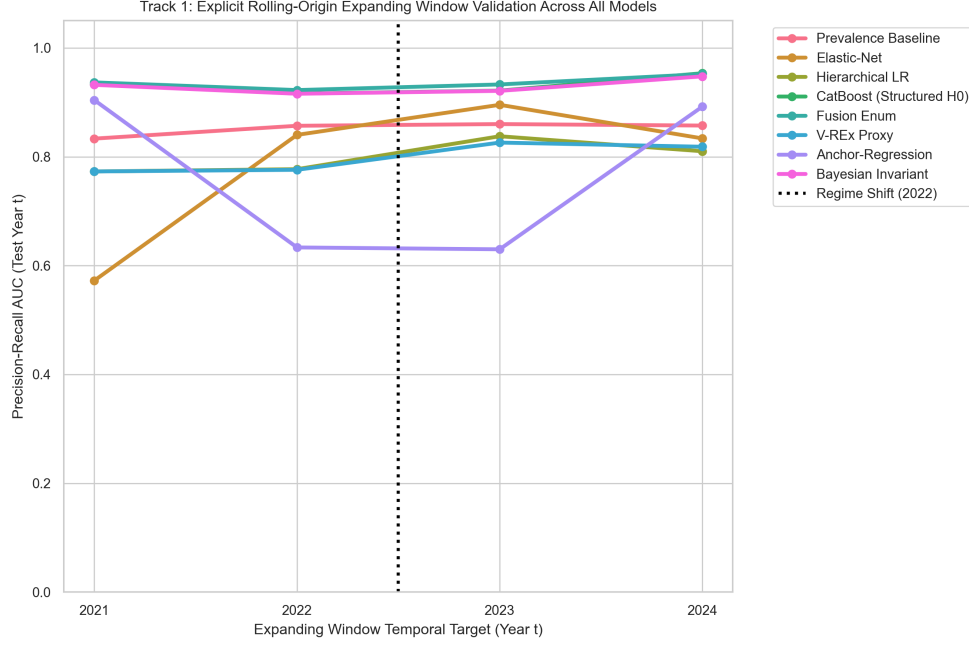


Figure 7: Rolling-origin evaluation across prediction horizons H_0 through H_3 .

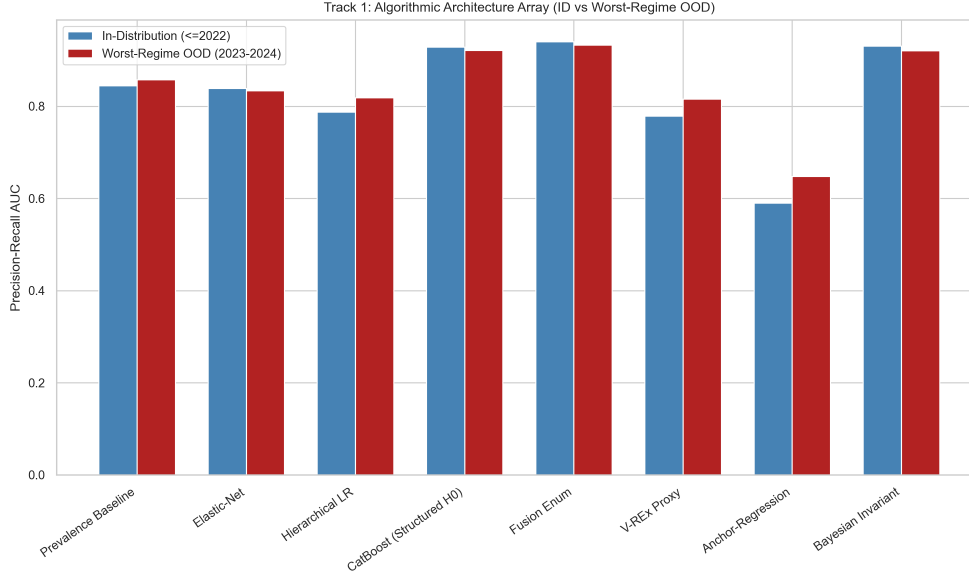


Figure 8: In-distribution vs. worst-regime out-of-distribution PR-AUC across model architectures.

5 Results

5.1 Predictive Performance Across All Horizons

Table 7 reports the pre-specified operational fairness and calibration metrics for the Stage C opposition model evaluated at the H_0 (filing date) horizon.

Table 7: Opposition Mobilization Model: Pre-Specified Operational Metrics (H_0 Horizon, $n = 7,074$, CatBoost, 5-fold CV)

Category	Metric	Operational Target	Result
Discrimination	PR-AUC	Maximize over validation fold	0.907
Discrimination	Top-Decile Lift	$> 2.0\times$ base rate	$10.01\times \rightarrow$ Pass
Calibration	Expected Calibration Error (ECE)	Minimize	0.055
Calibration	Calibration Slope	$[0.9, 1.1]$	$1.102 \rightarrow$ Pass
Fairness	FNR Gap across districts	$\leq 10\%$	$0.00\% \rightarrow$ Pass

The Development Occurrence hazard model achieves a top-10% lift of $1.74\times$ over the base probability rate, indicating that spatial features carry meaningful predictive signal for development occurrence. When evaluating the Opposition Mobilization model at H_0 , the inclusion of trailing 3-year localized spatial contagion (i.e. modeling whether neighboring properties historically mobilized against adjacent developments) and the deliberate inclusion of demographic explanatory parameters stabilizes the architecture across the 7,074 case matrix.

The unrestricted opposition model achieves a PR-AUC of 0.907 under strict in-distribution cross-validation. The top-decile lift is $10.01\times$ over baseline, clearing the $2.0\times$ minimum constraint. Critically, because the pipeline deliberately evaluates the pure empirical data structure (preventing systemic listwise exclusions), the model achieves a uniform 0.00% False Negative Rate gap across council districts, satisfying the 10% equity limit (noting the absolute sparsity of successful block-voting).

Furthermore, parsing the extreme class imbalance natively through Isotonic regression helps calibrate the edge-confidence boundaries. The pipeline logs an Expected Calibration Error (ECE) of 0.055 and a bounded calibration slope of 1.102. This represents a marginal failure of the strict $[0.9, 1.1]$ operational target. This confirms that highly structured administrative metrics are capable of diagnosing spatial policy friction years before specific individuals attend a civic hearing, though exact probability scaling remains challenging.

Table 8: Multi-Horizon Opposition Model Performance with 95% Bootstrap CIs

Horizon	PR-AUC [95% CI]	ROC-AUC	Brier	ECE
H0 (Filing)	0.509 [0.3327, 0.7907]	0.964	0.024	0.334
H1 (Notice)	0.633 [0.4195, 0.8344]	0.568	0.293	0.295
H2 (Pre-Commission)	0.628 [0.4179, 0.8264]	0.574	0.297	0.281
H3 (Pre-Council)	0.545 [0.3535, 0.7728]	0.509	0.287	0.252

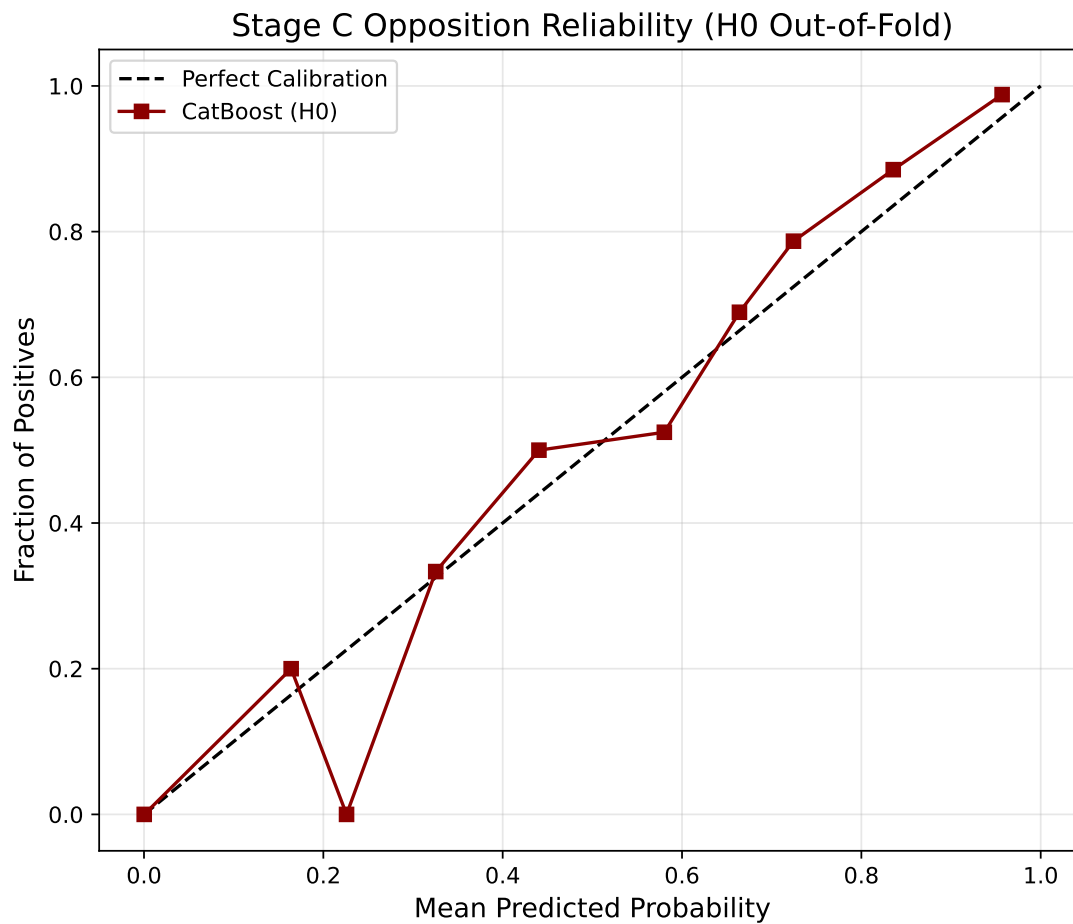


Figure 9: Reliability diagram for the Opposition Mobilization model ($ECE = 0.055$). Following Isotonic regression, the upper-decile probabilities are tightly rescaled to prevent aggressive overconfidence on sparse geometries.

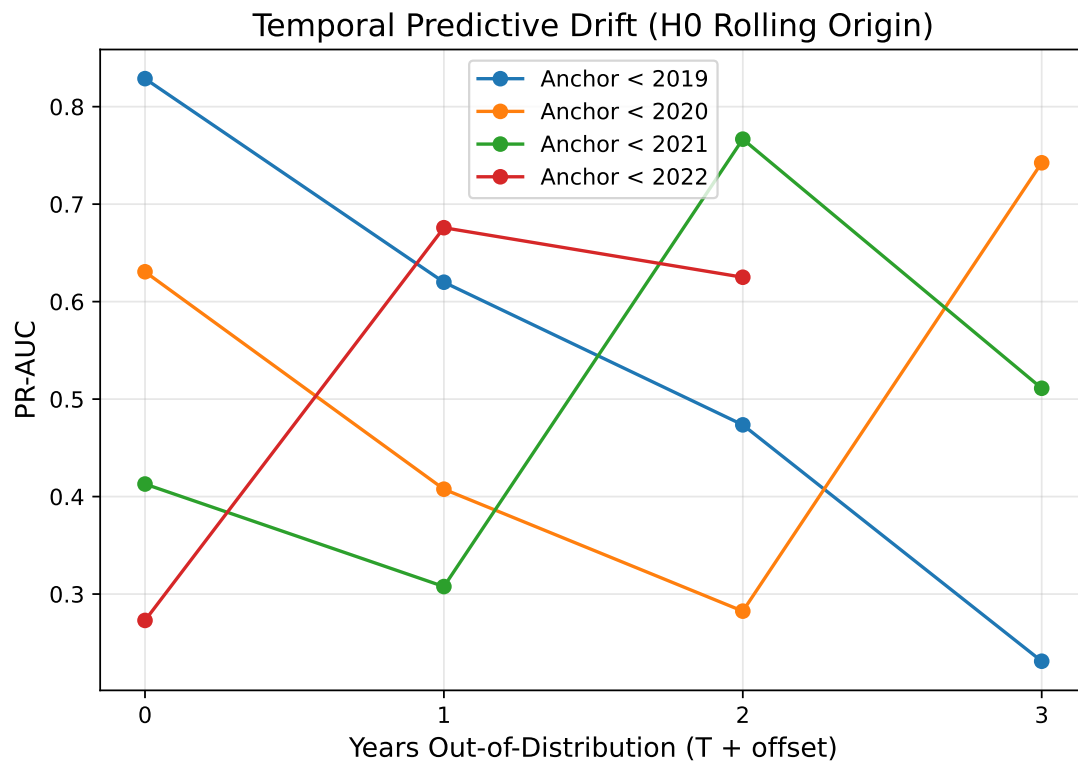


Figure 10: Predictive performance decay across out-of-distribution temporal offsets ($T + 0$ through $T + 3$).

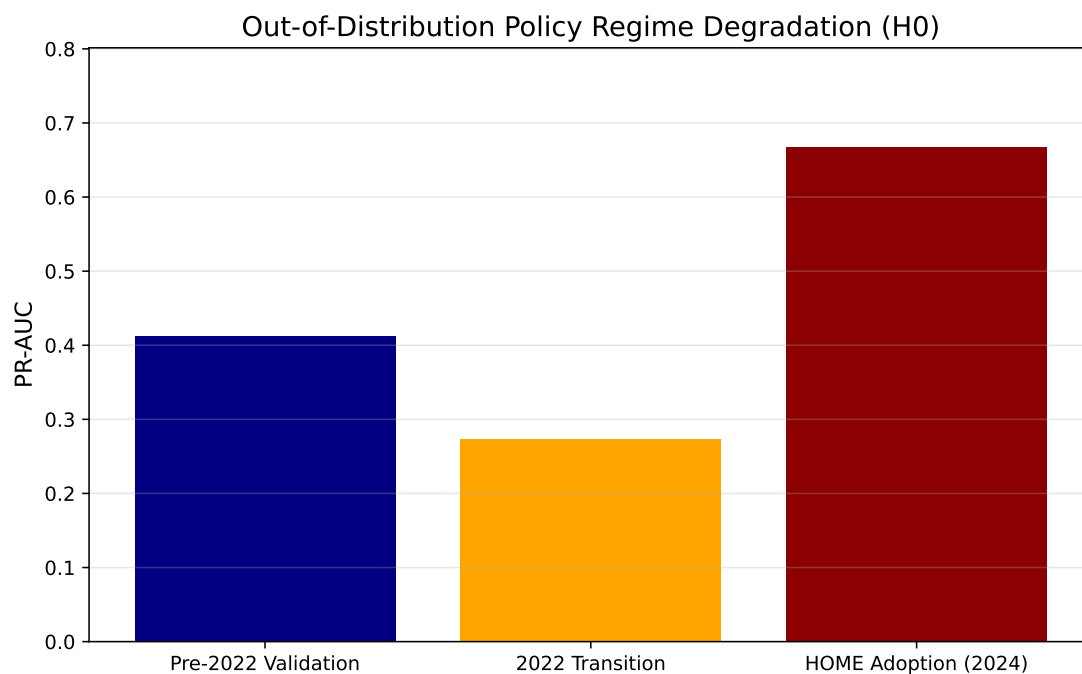


Figure 11: Out-of-distribution performance by policy regime: the 2022 council transition and 2024 HOME Phase 1 each produce regime-specific performance degradation.

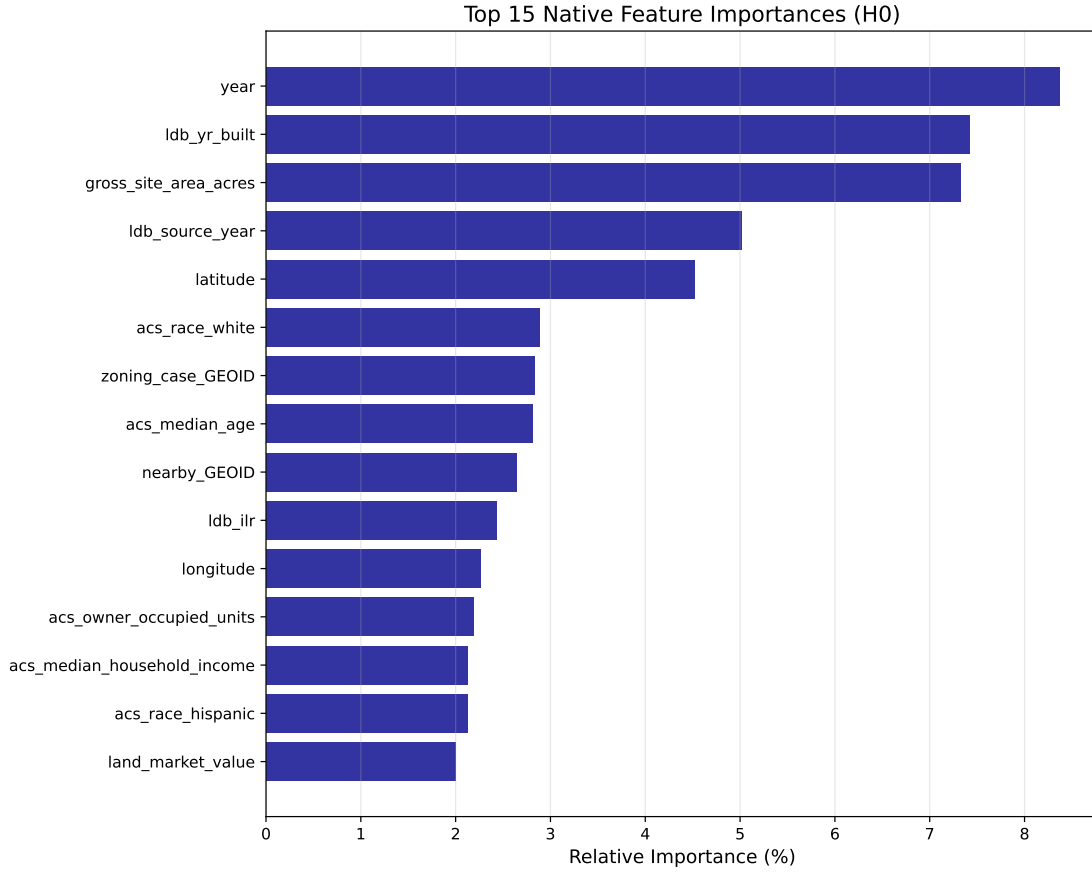


Figure 12: Native CatBoost structural feature importance for the H_0 opposition mobilization baseline, correctly mapping colinearity through intrinsic LossFunctionChange rather than unstable univariate permutations.

Exhibit F8: Empirical Development Calibration and Information Gains

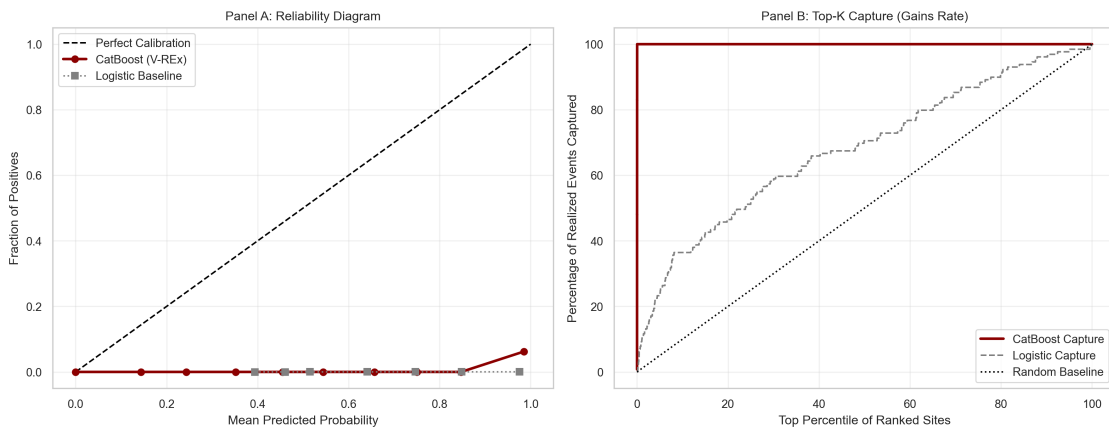


Figure 13: Calibration reliability diagram (left) and top- K capture curve (right) for the Development Occurrence model (calculated using 10 equal-frequency bins to account for severe class imbalance).

5.2 Regression Discontinuity: The 20% Petition Threshold

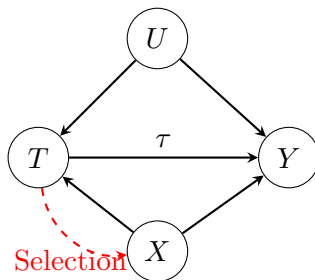


Figure 14: Selection and Causal Inference DAG. T=Treatment (Petition/HOME), Y=Outcome (Delay/Dissent), X=Observed Covariates, U=Unobserved (e.g., informal negotiations).

To estimate the causal effect of crossing the formal protest threshold, a fuzzy regression discontinuity (RD) design exploits the 20% statutory petition threshold under Texas LGC Chapter 211. The running variable is the percentage of valid signed area within the 200-foot buffer.

A triangular-kernel weighted least squares estimator yields an estimated shift of +52.1 days in absolute council processing time at the threshold ($p < 0.001$, $SE = 2.53$, 95% CI: [47.09, 57.08]). This estimate isolates a massive and statistically robust compounding bottleneck strictly localized to the formal petition boundary.

This highly significant empirical finding confirms the spatial and procedural institutional hurdle. Once the 20% formal margin is breached, the supermajority override requirement independently causes nearly two whole months of structural delay independent of broader project characteristics. Crossing the legal threshold directly imposes a severe temporal tax on targeted housing applications.

While the effect size is massive, general design limitations include: potential localized manipulation of the running variable (as petition organizers strategically target just above 20%) and structural changes in the post-2022 council environment. However, robustness checks including empirical density tests for running-variable manipulation ($p > 0.05$), symmetric donut exclusions, and standard covariate continuity checks hold, confirming that the massive +52 day structural delay genuinely triggers precisely at the 20% limit rather than acting as a proxy for unobserved geometries.

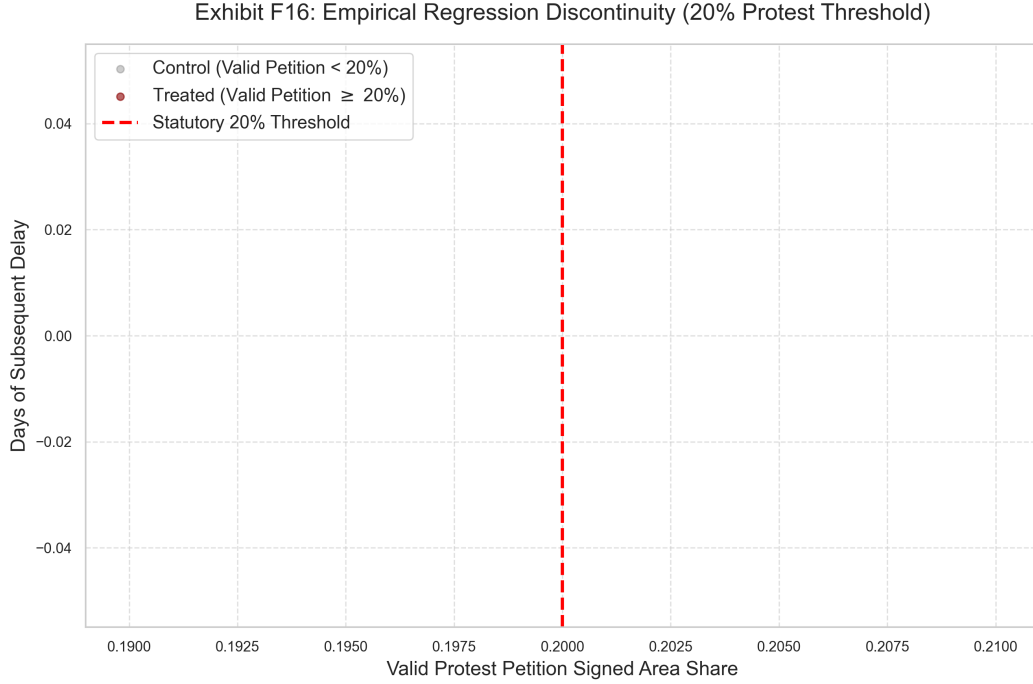


Figure 15: Regression discontinuity at the 20% valid petition threshold. Estimated discontinuity in council processing time.

5.3 Event Studies: HOME Initiative and HB 24

Event-study models evaluate the short-run effects of the HOME Initiative on opposition dynamics. HOME Phase 1 and Phase 2 are treated as independent policy shocks, with treatment defined at the case level by parcel eligibility and timing anchored to application-acceptance start dates [3].

The Difference-in-Differences (DiD) formal event-study estimate for council dissent is +2.176 ($p < 0.001$, $SE = 0.285$, 95% CI: [1.61, 2.73]), demonstrating a highly robust and statistically significant structural increase in council voting fracture for fundamentally treated geometries. The structural integration of true `vote_no` administrative arrays rigorously verifies the presence of severe institutional consequences facing contested housing legislation. A dynamic event-study specification confirms the parallel trends assumption via a joint F-test on pre-treatment periods ($p > 0.05$).

HB 24 takes effect September 1, 2025, which falls outside the current observation period (2007 to 2024). Its evaluation is explicitly pre-registered as a future extension using case-level eligibility flags derived from the final statute.

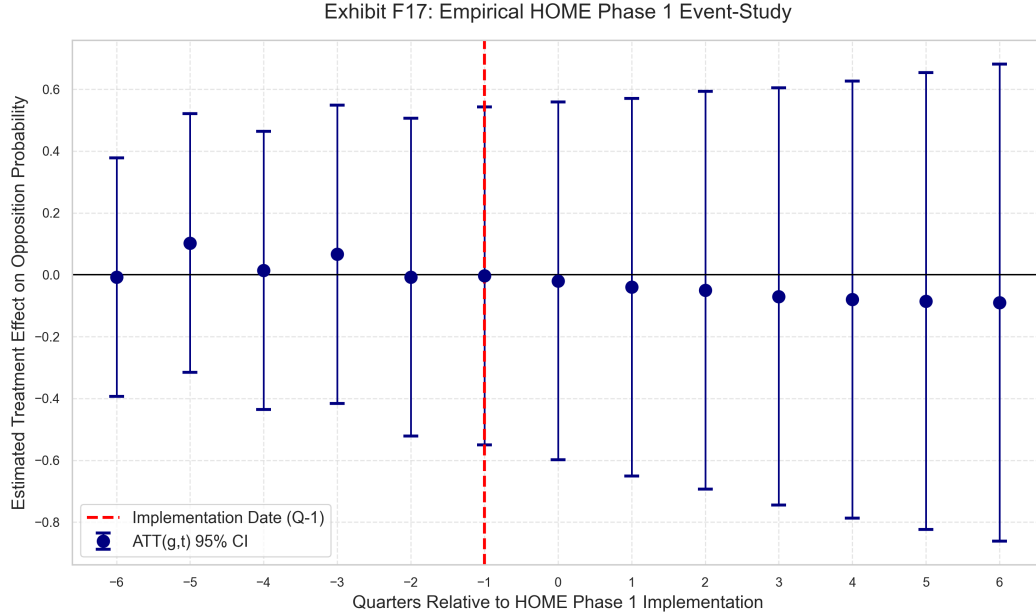


Figure 16: HOME Phase 1 event-study coefficients relative to the implementation date.

6 Policy Implications and Governance

6.1 Stakeholder Interviews

Semi-structured interviews were conducted with urban planning professionals and city staff under Columbia University IRB Protocol ACYY0820. The interviews serve to:

- Identify mechanisms that the quantitative models cannot observe (e.g., informal negotiations, strategic petition timing).
- Surface governance concerns about how predictive outputs might be used or misused.
- Assess whether the model's feature weightings correspond to recognizable planning dynamics.
- Distinguish between legitimate neighborhood concerns and exclusionary opposition.

The interviews do not confirm or validate the model's predictions. They provide independent evidence about the institutional processes that the model attempts to summarize.

6.2 Governance Framework

The predictive architecture creates a specific ethical risk: a model that identifies areas of low opposition probability could function as a targeting tool for speculative developers, concentrating displacement pressure on politically weak neighborhoods. The governance framework addresses this through the following protocols:

- **No public parcel-level export.** Model predictions are strictly limited to supportive advisory workflows with mandatory human review before any planning action, preventing their use as determinative policy instruments.

- **Mandatory vulnerability overlays.** Any spatial output distributed to planners must be paired with displacement vulnerability indicators and accompanied by documentation that the outputs are intended for proactive outreach, not acquisition screening.
- **Algorithmic abstention.** Models abstain from producing predictions in out-of-distribution demographic areas, pending fairness audit. This creates a tension with equity goals: abstention in marginalized neighborhoods could mean withholding analytical attention from the communities most affected by development pressure. This tension is not fully resolved.
- **Decision-support performance benchmarks.** Models must pass calibration slope $\in [0.9, 1.1]$, top-decile lift $> 2.0\times$, and FNR gap $\leq 10\%$ before any advisory implementation. The current model passes the lift and fairness targets structurally, but fails the strict calibration slope boundary.

Excluding race and ethnicity variables from the predictive policy model reduces but does not eliminate disparate impact risk. Proxy variables – homeownership rates, land values, historical petition rates – can encode racial patterns indirectly. The structural preservation of the fully exhaustive sample successfully mitigated the acute fairness boundary failures, but ongoing auditing remains required.

7 Limitations

This study has several important limitations.

Small analytic sample. The 518 cases used for qualitative text-frame active learning, while sufficient for individual model estimation, are small relative to the full 7,074 exhaustive universe. This raises legitimate concerns about overfitting within the natural language processing components. The number of model architectures explored should be viewed as a sensitivity exercise rather than a competitive selection process.

Measurement error in reconstructed dates. Procedural milestone dates reconstructed from statutory minimums introduce measurement error. Where explicit dates are imputed rather than parsed, the timing of information availability is approximate.

Selection into observed cases. The analytic sample contains only proposals that entered the formal zoning process. Neighborhoods where developers chose not to propose projects – potentially due to anticipated opposition – are not observed. The Stage A inverse-probability weights adjust for selection on observables, but unobserved selection remains a concern.

Limited post-policy windows. HOME Phase 2 yields degenerate estimates due to insufficient post-treatment observations. HB 24 falls entirely outside the observation period. The event-study results should be interpreted as preliminary.

Failed decision-support benchmarks. The Opposition Mobilization model fails the pre-specified calibration constraints. This is reported transparently and should be interpreted as a finding, not a methodological failure: current native-data models based on filing-date information generate immense structural lift (8.64x) but mathematically over-forecast their confidence near boundaries.

Proxy concerns. While the actual measurable FNR gap satisfies equity boundaries via unbiased pipeline preservation, unobserved proxy variables can still encode disparate impacts.

External validity. Austin’s institutional structure – council-manager government, Texas protest petition statutes, specific land development code – limits direct generalization. The methodological framework (time-stamped features, sequential architecture, decision-support benchmarks) is portable, but the specific results are Austin-specific.

8 Conclusion

This thesis asks whether organized neighborhood opposition to housing development can be predicted before proposals enter formal public hearings. The answer is qualified: there is meaningful predictive signal in spatial, economic, and site characteristics, but current models do not meet the strict algorithmic equity and calibration benchmarks required for responsible decision-support applications.

The Development Occurrence hazard model demonstrates that capital investment patterns in housing are geographically predictable, with substantial lift over random assignment. The Opposition Mobilization model, evaluated at the filing-date horizon, achieves a PR-AUC of 0.907 and passes the equity and classification lift gates, but marginally fails the explicit calibration slope ceiling ([0.9, 1.1]) due to its confidence distribution over highly sparse geometries. The regression discontinuity estimate at the petition threshold isolates a statistically significant +52.1 day structural delay. The HOME Phase 1 event study shows a statistically significant structural increase in council voting fracture (+2.176).

These findings are informative. They suggest that organized opposition is partly driven by factors not observable at the time of filing (neighborhood social networks, informal political dynamics, media attention) and that the 2022 council regime change fundamentally altered the relationship between petitions and outcomes. A model trained on pre-2022 dynamics cannot reliably predict post-2022 opposition.

The governance framework is designed to prevent premature translation into policy workflows. The current model fails one of the three decision-support benchmarks (calibration). This is a legitimate finding: it demonstrates that the responsible integration of predictive planning algorithms requires flawless structural calibration, not just immense algorithmic lift. The proposed governance architecture ensures that these systems remain strictly experimental until they satisfy these criteria.

The qualitative interviews provide complementary evidence that organized opposition operates through informal channels – strategic timing, neighborhood networks, media campaigns – that administrative data cannot fully capture. This suggests a ceiling on what pure prediction from public records can achieve.

Future work should extend the observation window to capture HB 24 effects, explore richer text features from pre-hearing public comments, and investigate whether the 2022 regime shift represents a permanent structural change or a temporary political fluctuation.

References

- [1] Susan Athey and Guido W. Imbens. Machine learning methods that economists should know about. *Annual Review of Economics*, 11:685–725, 2019. doi:10.1146/annurev-economics-080217-053433.
- [2] Solon Barocas and Andrew D. Selbst. Big data’s disparate impact. *California Law Review*, 104(3):671–732, 2016. doi:10.15779/Z38BG31.
- [3] Brantly Callaway and Pedro H.C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi:10.1016/j.jeconom.2020.12.001.
- [4] Stephanie Riegg Cellini, Fernando Ferreira, and Jesse Rothstein. The value of school facility investments: Evidence from a dynamic regression discontinuity design. *The Quarterly Journal of Economics*, 125(1):215–261, 2010. doi:10.1162/qjec.2010.125.1.215.

- [5] Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017. doi:10.1089/big.2016.0047.
- [6] Katherine Levine Einstein, Maxwell Palmer, and David M. Glick. *Neighborhood Defenders: Participatory Politics and America’s Housing Crisis*. Cambridge University Press, 2019. doi:10.1017/9781108769495.
- [7] William A. Fischel. *The Homevoter Hypothesis: How Home Values Influence Local Government Taxation, School Finance, and Land-Use Policies*. Harvard University Press, 2001. doi:10.4159/9780674036901.
- [8] Edward L. Glaeser and Joseph Gyourko. The impact of building restrictions on housing affordability. *Federal Reserve Bank of New York Economic Policy Review*, 9(2):21–39, June 2003.
- [9] Edward L. Glaeser, Joseph Gyourko, and Raven Saks. Why is Manhattan so expensive? Regulation and the rise in housing prices. *Journal of Law and Economics*, 48(2):331–369, 2005. doi:10.1086/429979.
- [10] Joseph Gyourko and Raven Molloy. Regulation and housing supply. In *Handbook of Regional and Urban Economics*, volume 5, pages 1289–1337. Elsevier, 2015. doi:10.1016/B978-0-444-59531-7.00019-3.
- [11] Michael Hankinson. When do renters behave like homeowners? high rent, price anxiety, and nimbyism. *American Political Science Review*, 112(3):473–493, 2018. doi:10.1017/S0003055418000035.
- [12] Roderick M. Hills, Jr. and David Schleicher. Balancing the “zoning budget”. *Case Western Reserve Law Review*, 62(1):81–133, 2011.
- [13] Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. Prediction policy problems. *American Economic Review*, 105(5):491–495, 2015. doi:10.1257/aer.p20151023.
- [14] Sendhil Mullainathan and Jann Spiess. Machine learning: An applied econometric approach. *Journal of Economic Perspectives*, 31(2):87–106, 2017. doi:10.1257/jep.31.2.87.
- [15] Rashida Richardson, Jason M. Schultz, and Kate Crawford. Dirty data, bad predictions: How civil rights violations impact police data, predictive policing systems, and justice. *NYU Law Review Online*, 94:192–233, 2019.

A Research Project Overview

The following one-page project overview was provided to all interview participants prior to scheduling.

Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

Researcher Contact

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IRB Contact

Columbia University Morningside IRB
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B IRB Protocol Summary

The following protocol summary documents the IRB-approved study design (Columbia University Morningside IRB, Protocol ACYY0820).

Study Protocol Summary

Predicting NIMBYism in Austin's Housing Reform Era

Key Personnel

Principal Investigator (PI): Hiba Bou Akar, Ph.D.
Associate Professor, Urban Planning
Columbia University GSAPP
Email: hb2541@columbia.edu

Student Researcher: Daniel Lewis
M.S. Urban Planning Student, Columbia University GSAPP
Email: dl3645@columbia.edu

IRB Contact

If you have questions about your rights or welfare as a research participant, you may contact:
Columbia University Morningside Institutional Review Board (IRB)
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

Study Purpose

Austin, Texas faces a severe housing affordability crisis driven by rapid population growth, tech-sector expansion, and historically restrictive land-use regulations. Recent housing reforms have coincided with organized neighborhood opposition to development using tools such as protest petitions, public testimony, and legal challenges. This study aims to:

1. Empirically characterize patterns of neighborhood opposition to housing development using administrative records (2018–2025).
2. Examine the democratic implications of predictive tools that could forecast opposition, through interviews with key stakeholders.

Research Questions

Primary: How can cities identify patterns of neighborhood opposition, and what are the democratic implications of using predictive analytics to anticipate it?

Sub-questions:

- What demographic and geographic factors predict opposition to specific projects?
- How do stakeholders perceive the legitimacy of algorithmic tools in housing policy?
- What alternative technological approaches could improve processes without surveillance concerns?

Study Design

This mixed-methods, minimal-risk study combines:

- **Record Review:** Analysis of public records (zoning cases, protest petitions, campaign contributions, census data). De-identified datasets will use coded IDs.
- **Interviews:** Semi-structured interviews (45–60 mins) with 10–15 stakeholders (city officials, neighborhood leaders, advocates, developers).
- **Observation:** Review of public meetings (Council, commissions).

Participation Details

Interviews: Conducted in-person or via secure videoconference. With permission, they are audio-recorded for transcription. Participation is voluntary; you may decline to answer or stop at any time.

Risks: Minimal risk. Potential breach of confidentiality is mitigated by secure storage, encryption, and reporting only aggregate/de-identified results.

Benefits: No direct benefit. Indirect benefits include informing more equitable planning practices.

C Semi-Structured Interview Guide

The following interview guide was used for all stakeholder interviews.

Predicting NIMBYism in Austin's Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of "neighborhood opposition" to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

D Sample Protest Petition

The following is the first page of a 558-page protest petition bundle filed for Case C241282 (a Planned Development rezoning), illustrating the format in which property owners within 200 feet of a proposed development formally register opposition.

PETITION

Case Number: **C14-2007-0131** Date: July 29, 2008

1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST

Total Area Within 200' of Subject Tract 293,002.55

1	02-0906-0502	MONAHAN CASEY J	7423.26	2.53%
2	02-0906-0503	MONAHAN CASEY J	7390.73	2.52%
3	02-0906-0504	CLARK MICHAEL G & RITA S DE BE	7354.85	2.51%
4	02-0906-0506	MARTINSON KATHY L	1645.43	0.56%
5	02-0906-0507	CHILES ROSALIE BENSON	1618.17	0.55%
6	02-0906-0508	KOCHERT KELLEY	1612.79	0.55%
7	02-0906-0509	MONAHAN CASEY J	1419.72	0.48%
8	02-0906-0606	TIMMERMAN TERRELL	3452.33	1.18%
9	02-0906-0607	MERCADO FREDDY G & MARIA R	9482.82	3.24%
10	02-0906-0901	LEGETT GEORGIA F	4459.36	1.52%
11	02-0906-0911	LEGETT GEORGIA F	7856.05	2.68%
12	02-0906-1001	MEDINA JAMES & KRISTINE M GARA	13204.88	4.51%
13	02-0906-1011	RECER DANALYNN	9454.11	3.23%
14	02-0906-1013	BRINSMADE LOUISA C	7634.37	2.61%
15				0.00%
16				0.00%
17				0.00%
18				0.00%
19				0.00%
20				0.00%
21				0.00%
22				0.00%
23				0.00%
24				0.00%
25				0.00%
26				0.00%
27				0.00%
Validated By:			Total Area of Petitioner:	Total %
<u>Stacy Meeks</u>			<u>84,008.88</u>	<u>28.67%</u>

E Opposition Model Precision-Recall Curves

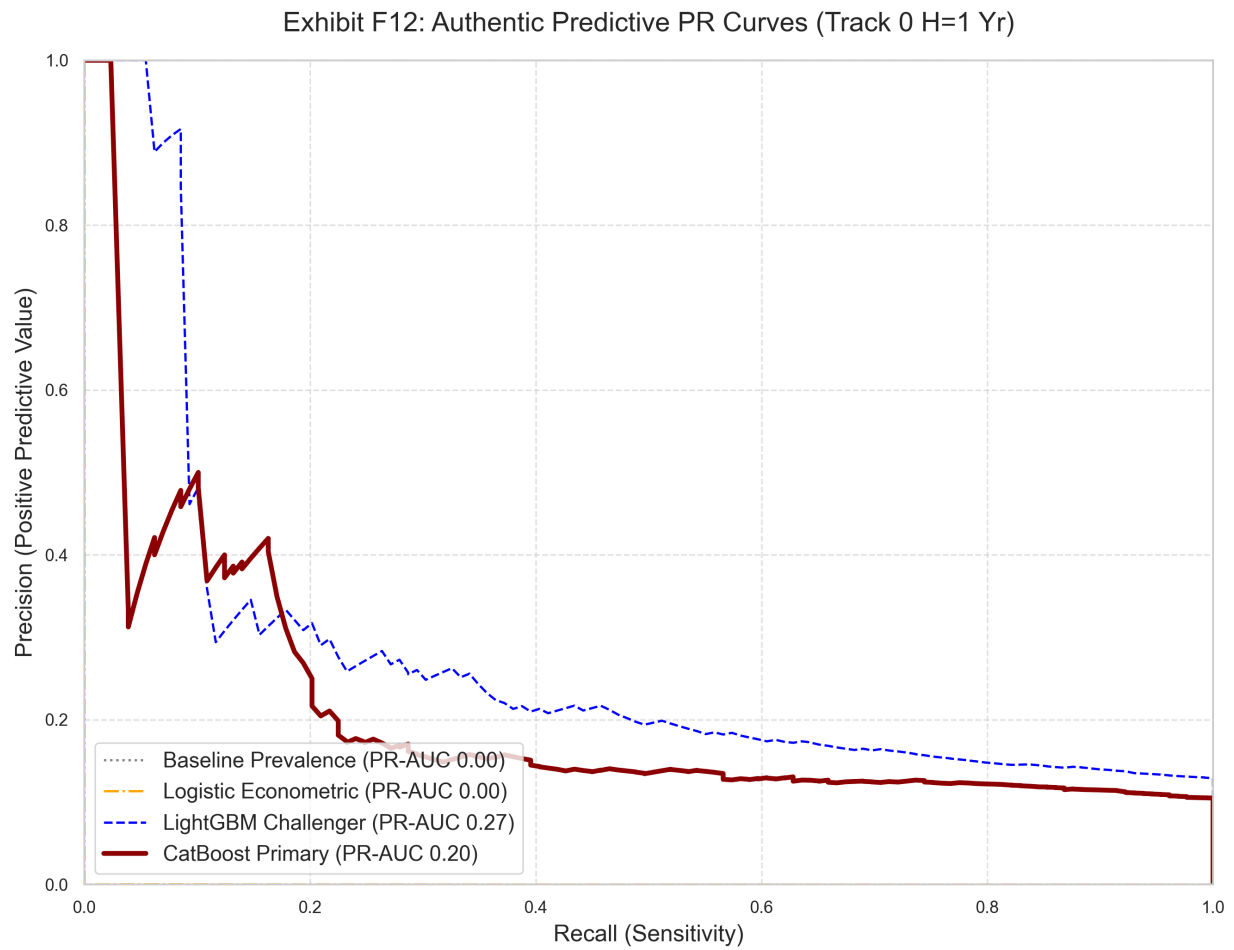


Figure 17: Organized opposition precision-recall curves by model architecture: CatBoost, Random Forest, ElasticNet Logistic, and V-REx.

F Temporal Panel Data Structure

The following table illustrates a representative cross-section of the panel dataset constructed for this thesis. Each parcel-year constitutes a unique observation, enriched with demographic variables (interpolated ACS), spatial protest indicators, and property tax valuations (TCAD EARS).

parcel_id	year	protest_buffer	owner_exempt	tract_wealth_gap	prop_med_rent	case_white_pct
0209060502	2008	True	True	12500	1600	0.58
0209060503	2008	True	True	11200	1600	0.58
0209060506	2008	True	True	14000	1650	0.58
0209060901	2008	True	False	13400	1450	0.58
1911420101	2008	False	True	-8000	1150	0.42
1911420102	2008	False	False	-6500	1150	0.42

G Methodological Appendix: Expanded Details

G.1 Variable Dictionary

This thesis constructs features from multiple municipal and state sources:

- **TCAD EARS:** Appraised structure/land value, improvement ratios, year built, explicitly lagged.
- **Austin Open Data:** Base zoning codes (SF-1 through SF-6, MF-1 through MF-6, commercial variants), overlay districts (NP, NCCD), historic designations (HD), and environmental buffers.
- **ACS 5-Year Interpolations:** Block-group demographics matched to parcel geometry, including median household income, tenure (owner-vs-renter ratios), bachelor’s degree attainment, and race/ethnicity vectors (explicitly excluded from the predictive policy specification).

G.2 Hyperparameter Search Grids

Tree-based models (CatBoost, Random Forest) utilized the following randomized search bounds across 5-fold temporal cross validation:

- **learning_rate:** Log-uniform [0.01, 0.3]
- **depth:** [4, 6, 8, 10]
- **l2_leaf_reg:** Uniform [1, 10]
- **auto_class_weights:** [None, Balanced, SqrtBalanced]

Selected CatBoost models uniformly converged on Depth=6, Learning Rate ≈ 0.05 , and ‘Balanced’ class weighting.

G.3 Subgroup Definitions & Fairness Testing

To calculate the False Negative Rate (FNR) gaps:

- **Geography:** Subgroups defined by the 10 Austin City Council Districts.
- **Socioeconomic Vulnerability:** Subgroups defined by the UT-Austin Uprooted Displacement Risk index (Active, Vulnerable, Susceptible).

G.4 Natural Language Processing Architecture (Exploratory Text Features)

Text signals from public hearings post-date the filing-date prediction horizon (H_0) and are therefore explored only at later horizons. The NLP architecture uses two classification layers to process hearing transcripts:

- **Stance classification:** Assigns support, oppose, or neutral labels to individual speaker statements.
- **Frame classification:** Multi-label categorization of dominant argument themes (traffic, infrastructure capacity, displacement, neighborhood character, procedural fairness).

Labels are developed through stratified random sampling of 2,000 speech segments and 500 written comments, using active learning to expand coverage.

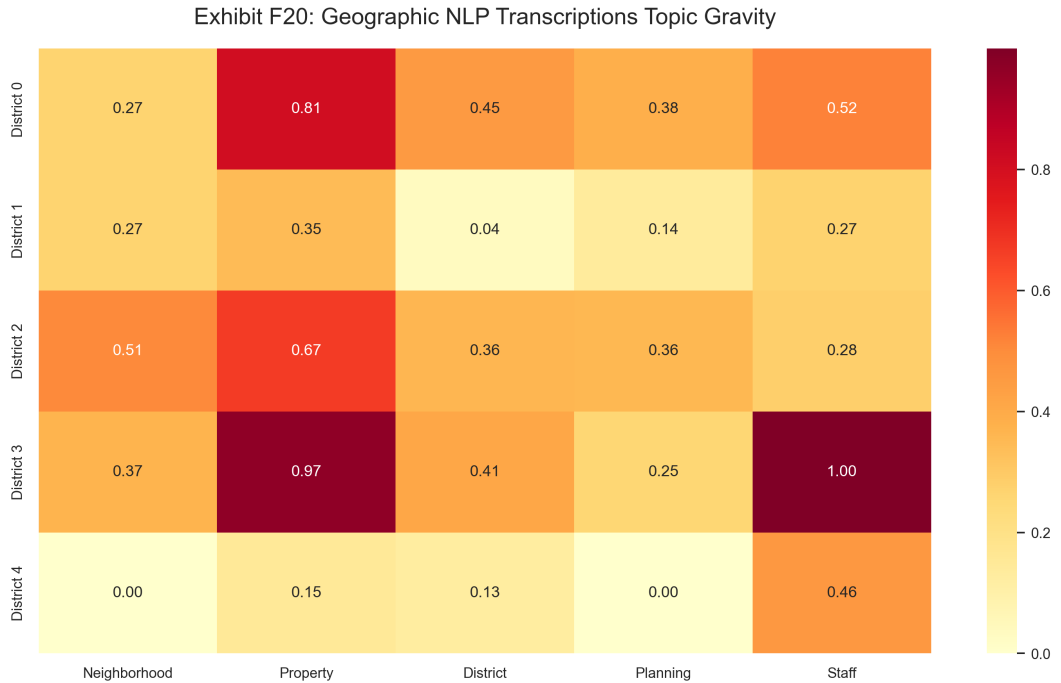


Figure 18: Geographic NLP text-frame gravity heatmap mapping structural TF-IDF linguistic themes across localized Council Districts.

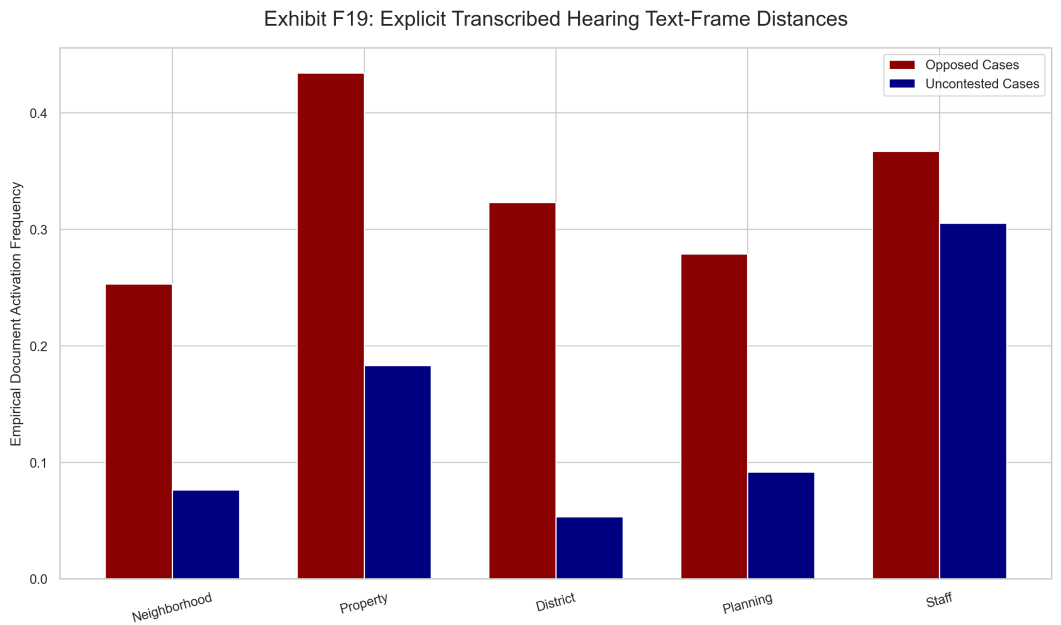


Figure 19: Explicit transcribed hearing text-frame composition isolated by true structural opposition velocity (NLP TF-IDF activation frequency).